

Week 2: LCM (Least Common Multiple)

JSS1 Mathematics — First Term Lesson Note (Deep-Dive Edition)

Introduction

LCM helps us solve problems involving repeating events or combining quantities with different groupings — for example, working out when two market days that occur every 4 and 6 days will fall on the same day again.

Definition

The **LCM** of two or more numbers is the smallest number that is a multiple of all of them.

Method 1 — Listing Multiples

Example 1: Find the LCM of 4 and 6.

Multiples of 4: 4, 8, 12, 16, 20, 24...

Multiples of 6: 6, 12, 18, 24, 30...

Common multiples: 12, 24...

LCM = 12 (the smallest common one)

Example 2: Find the LCM of 5 and 8.

Multiples of 5: 5, 10, 15, 20, 25, 30, 35, 40...

Multiples of 8: 8, 16, 24, 32, 40...

Common multiples: 40...

LCM = 40

Method 2 — Prime Factorization

Example 1: Find the LCM of 18 and 24.

$$18 = 2 \times 3 \times 3 = 2 \times 3^2$$

$$24 = 2 \times 2 \times 2 \times 3 = 2^3 \times 3$$

Take the **highest power** of each prime factor: $2^3 \times 3^2 = 8 \times 9 = 72$

LCM = 72

Example 2: Find the LCM of 10, 15, and 20.

$$10 = 2 \times 5$$

$$15 = 3 \times 5$$

$$20 = 2^2 \times 5$$

$$\text{LCM} = 2^2 \times 3 \times 5 = 4 \times 3 \times 5 = \mathbf{60}$$

Real-Life Application

A school bell rings for break every 20 minutes and the water tanker arrives every 30 minutes. The LCM of 20 and 30 is 60, meaning both events happen together every 60 minutes (once an hour). LCM is also used when adding fractions with different denominators.

Vocabulary

Word	Meaning
Multiple	The result of multiplying a number by a whole number (e.g., multiples of 3: 3, 6, 9...)
Prime number	A number greater than 1 with exactly two factors: 1 and itself (e.g., 2, 3, 5, 7, 11)
Prime factorization	Writing a number as a product of only prime numbers

Evaluation Questions

- 1 Find the LCM of 8 and 12 by listing multiples.
- 2 Find the LCM of 15 and 25 using prime factorization.
- 3 Find the LCM of 6, 9, and 12.
- 4 Find the LCM of 7 and 9.
- 5 Explain, in your own words, what "LCM" means.
- 6 Two bells ring every 15 minutes and every 25 minutes. After how many minutes will they ring together?
- 7 Find the LCM of 12, 16, and 24 using prime factorization.
- 8 Why is the LCM never smaller than the largest of the given numbers?
- 9 Find the LCM of 3 and 5. What do you notice about the answer when two numbers share no common factors?
- 10 List the first five multiples of 9.